WF-Mamba: Weighted Fourier State-Space Models With Adaptive Spatial-Frequency Synergy for Hyperspectral Unmixing

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***Abstract*—Hyperspectral unmixing (HU) is a crucial technique that aims to estimate subpixel-level ground object information in remote sensing images. Existing deep learning-based unmixing methods, especially autoencoders (AEs), typically operate only within the original spatial–spectral feature domain. However, due to the impact of noise and spectral variability, relying solely on the original domain makes it difficult to fully extract effective features and leaves the models highly susceptible to interference. To address these issues, we present an innovative dual-stream spatial-frequency state-space model (SSM), termed WF-Mamba, for enhanced HU. A physics-driven snake-like scanning (SS) mechanism is introduced in WF-Mamba to remodel the data serialization process, which strictly preserves the intrinsic neighborhood contiguity and 2-D topological priors. To extend the feature domain and overcome the frequency-domain blindness of pure spatial models, a macroscopic weighted Fourier neural operator (WFNO) branch is constructed. By dynamically recalibrating global low-frequency modes in the complex domain, WFNO functions as an advanced physical filter to actively suppress high-frequency noise. Furthermore, an adaptive synergistic fusion (ASF) mechanism is proposed for WF-Mamba to break the bottleneck of rigid feature concatenation. More robust features can be obtained by ASF by autonomously balancing the sharp local spatial boundaries extracted by Mamba with the globally purified smoothness from WFNO via dynamic residual gating. The fused robust features are then mapped to estimate accurate abundances and reconstruct the image. Extensive experiments on one synthetic and three real hyperspectral datasets demonstrate that the proposed WF-Mamba achieves competitive performance compared to advanced methods.**

***Index Terms*—Adaptive synergistic fusion (ASF), Fourier neural operator (FNO), hyperspectral unmixing (HU), state-space model (SSM).**

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I. Introduction

HYPERSPECTRAL images (HSIs) feature high resolution and the simultaneous acquisition of spatial and spectral information, providing a wealth of information for material identification. This unique capability has established HSI as a foundational tool in diverse remote sensing applications, ranging from ecological monitoring and mineral exploration to precision agriculture [1], [2]. However, due to the inherent tradeoff between spatial and spectral resolution in imaging sensors, the mixed-pixel phenomenon remains a ubiquitous challenge. In real-world scenarios, a single pixel often encompasses spectral contributions from multiple distinct materials [3], which can limit the performance of downstream tasks such as subpixel target detection and fine-grained classification. Hyperspectral unmixing (HU) is an effective method for handling mixed pixels. From a subpixel perspective, the goal of HU is to obtain the pure spectral signals (i.e., endmembers) within a pixel and analyze the ratio of each class of endmembers (i.e., abundances) within the pixel [4], [5].

Early research efforts primarily focused on traditional mathematical models, encompassing statistical, sparse regression, geometrical, and other various theoretical frameworks that laid a solid foundation for the field [1]. For instance, Themelis et al. [6] proposed a hierarchical Bayesian approach implementing sparsity-enforcing priors to model statistical spectral mixtures, effectively enhancing the robustness of endmember estimation. Iordache et al. [7] introduced collaborative sparse regression solved via alternating direction frameworks, which exploited library-based sparsity to improve unmixing accuracy. Furthermore, Lu et al. [8] developed a manifold-regularized nonnegative matrix factorization (NMF) algorithm integrating a graph Laplacian matrix, demonstrating certain advantages in preserving low-dimensional topological structures, a concept

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further validated by various constrained NMF frameworks [9]. While these traditional mathematical models offer strong interpretability and theoretical rigor, their reliance on simplified or linear assumptions makes it difficult for them to fully capture the complex, highly nonlinear latent representations inherent in real-world hyperspectral data.

To overcome the inherent limitations of linear assumptions and better model these complex nonlinearities, the unmixing paradigm has gradually shifted toward sophisticated deep learning architectures over the past decade. Particularly, autoencoder (AE)-based networks have risen to prominence due to their powerful representation capabilities. In a typical AE framework, the encoder directly estimates the abundance maps, while the decoder reconstructs the original image using weights that correspond to the endmember matrix. Researchers have extensively explored various AE structures to capture these complex nonlinear mappings, moving from basic deep fully connected neural networks [4] that proved the feasibility of deep feature extraction, to advanced sparse and stacked frameworks [10], [11] which effectively mitigated initialization sensitivities by combining structural priors with AE initialization. In parallel, multitask models like the multitask HU model (MAHUM) [12] have proven useful in joint abundance estimation and endmember optimization. Moreover, to address complex physical realities in remote sensing environments, advanced models have been developed to explicitly tackle spectral variability [13], [14], [15] and nonlinear mixing effects [16], [17], [18], thereby improving the physical reliability of the unmixing results.

Despite the success of these AE-based models in extracting complex spectral representations, they primarily treat individual pixels as isolated entities, often overlooking the structural correlations between adjacent pixels. Seeking to exploit the rich spatial context inherent in HSIs, researchers have widely integrated convolutional neural networks (CNNs) into the unmixing process. For example, Zhang et al. [19] developed a deep convolutional framework utilizing 2-D convolutional layers for localized spatial feature extraction, while Zhao et al. [20] introduced 3-D-CNN models applying 3-D convolutional kernels to simultaneously capture spectral and spatial correlations directly within data cubes. These methods have shown effectiveness in enforcing piecewise smoothness and preserving local spatial continuity in abundance maps [21]. To further refine these deep learning frameworks, abundance-guided attention networks [22] and multimodal insights [23] have been integrated to dynamically weight features, while automated neural architecture search (AutoNAS) [24] has been employed to optimize hyperparameter selection. However, CNN-based methods are primarily designed around the localized nature of convolutional kernels, meaning the receptive field expands gradually with network depth. Consequently, CNNs struggle to effectively model long-range global contexts, such as identifying identical materials distributed distantly across a large scene.

Over the past few years, to overcome the inherent localized limitations of convolutions, Vision Transformers (ViTs) have been increasingly adopted to establish explicit scene-wide dependencies [25]. For instance, Ghosh et al. [26] applied a spatial–spectral Transformer incorporating multihead self-attention along both spatial patches and spectral dimensions, achieving promising performance in global correlation modeling. Similarly, Duan et al. [27] designed an attention-driven network using tokenized spectral representations to effectively preserve global structural relationships. Nevertheless, the quadratic computational complexity $\mathcal{O}(N^2)$ of the standard self-attention mechanism relative to the sequence length presents distinct computational challenges, particularly when processing high-dimensional HSI data with large spatial resolutions [28].

Recently, state-space models (SSMs), particularly the Mamba architecture, have emerged as a compelling paradigm for sequence modeling [29]. In the field of HU, Chen et al. [30] introduced UNMamba, demonstrating the potential of cascaded spatial–spectral Mamba architectures to maintain competitive unmixing accuracy with reduced computational overhead. Meanwhile, Mamba has also been widely applied to other HSI processing tasks, including HSI classification and cross-scene domain adaptation [31]. For example, Liang et al. [32] introduced spatial and spectral Mamba blocks to jointly model long-range spatial–spectral dependencies, while SpectralMamba [33] explored efficient state-space modeling and spectral sequence scanning for hyperspectral data. These studies demonstrate the effectiveness of Mamba in capturing spatial–spectral or spectral dependencies. Despite its efficiency, the direct application of standard 1-D Mamba frameworks to 2-D HSI still reveals significant structural challenges [34], [35]. Specifically, conventional raster scanning introduces spatial discontinuities at row boundaries and disrupts the neighborhood continuity of geometric structures.

Furthermore, high-resolution HSI is inevitably affected by complex spatial noise distributions and atmospheric interference [36], which often manifest as prominent high-frequency artifacts. As revealed by studies from a frequency perspective, models operating exclusively in the spatial domain are highly vulnerable to high-frequency corruptions, severely compromising their structural robustness [37], [38], [39]. To overcome this frequency-domain blindness, Li et al. [40] pioneered the Fourier neural operator (FNO), which effectively captures global frequency representations and addresses the limitations of local receptive fields. However, standard FNOs typically rely on a static, globally shared weight tensor that treats all frequency modes uniformly. This rigid filtering mechanism lacks the dynamic weighting capacity required to adaptively recalibrate specific frequency modes in response to spatially varying HSI noise. Finally, while some existing multiview networks attempt to incorporate frequency information, they often rely on rigid feature concatenation. This static fusion approach may not dynamically synergize local spatial topological boundaries with global frequency-domain smoothness, limiting the model's adaptability to diverse noise levels across different spectral bands.

In light of the above challenges, we propose WF-Mamba, a novel dual-stream spatial-frequency SSM specifically tailored for HU. Our core philosophy is to integrate microscopic spatial sequence inference with macroscopic global frequency purification within a unified, adaptive synergistic framework [41]. To maintain the smoothness of land cover distributions during sequence modeling, we replace standard raster scanning with

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an adjacency-preserving snake-like scanning (SS) mechanism that preserves local continuity. Simultaneously, inspired by the recent success of FNOs [42], we introduce a parallel weighted FNO (WFNO) branch that dynamically recalibrates global frequency modes. As a natural physical low-pass and bandpass filter, the WFNO branch effectively suppresses high-frequency noise and enforces global spatial consistency. Finally, an adaptive residual fusion mechanism is designed to dynamically merge the spatial and frequency streams. This allows the network to intelligently balance spatial boundary preservation and frequency noise suppression based on the distinct data characteristics of different wavelet streams.

The main contributions of this article are summarized as follows.

1) We propose an adjacency-preserving SS mechanism to fundamentally remodel the serialization process of hyperspectral data. By alternating the traversal trajectory, this strategy significantly mitigates the spatial fragmentation caused by conventional raster scanning, thereby better preserving the intrinsic neighborhood contiguity and 2-D topological priors essential for high-fidelity abundance estimation.
2) To overcome the inherent frequency-domain blindness of pure spatial SSMs, we construct a macroscopic branch driven by the WFNO. By dynamically recalibrating global low-frequency modes in the complex domain, this operator functions as an advanced filter that actively suppresses pervasive high-frequency environmental noise and enforces global spectral–spatial consistency.
3) We develop a noise-robust adaptive spatial-frequency synergistic fusion mechanism that breaks the bottleneck of rigid feature concatenation. Through dynamic residual gating, this mechanism autonomously regulates the contribution of each domain, elegantly balancing the sharp local spatial boundaries extracted by the Mamba branch with the globally purified smoothness from the WFNO branch to achieve optimal anti-noise robustness.

The remainder of this article is organized as follows. Section II reviews pertinent prior work related to deep learning-based HU, SSMs, and frequency-domain analysis. Section III details the proposed WF-Mamba architecture, encompassing the multistream wavelet AE, the core visual Mamba layer, and the physics-constrained optimization objective. Section IV presents comprehensive experimental results, ablation studies, and discussions on both synthetic and real-world datasets. Finally, Section V concludes this article.

II. Related Works

Given that the proposed WF-Mamba framework relies on deep AEs, the linear mixing model (LMM), SSMs, and frequency-domain neural operators, a concise overview of their fundamental mathematical formulations and pertinent studies is provided in this section.

A. AE-Based LMM Unmixing

The LMM presupposes that the spectral signature of a mixed pixel can be viewed as a linear blend of pure endmember spectra. Mathematically, for a hyperspectral pixel $y \in \mathbb{R}^{L\times 1}$ with L spectral bands, it is formulated as

$$\mathbf{y} = \mathbf{M}\mathbf{a} + \mathbf{n} \tag{1}$$

where $M \in \mathbb{R}^{L\times R}$ denotes the endmember matrix containing R materials, $a \in \mathbb{R}^{R\times 1}$ represents the corresponding fractional abundances, and n is the additive noise.

Under the deep AE framework, the unmixing process naturally maps the LMM into an encoder–decoder architecture. The encoder function $f_\theta(\cdot)$ extracts the low-dimensional abundance representation $a = f_\theta(y)$, and the linear decoder reconstructs the pixel as $\hat{y} = Wa$. The weight matrix W directly corresponds to the endmember matrix M. To further exploit the spatial correlation of HSIs, CNNs have been widely integrated. The feature map $\mathcal{H}^{(l)}$ at the lth layer is typically obtained by

$$\mathcal{H}^{(l)} = \sigma\left(W^{(l)} * \mathcal{H}^{(l-1)} + b^{(l)}\right) \tag{2}$$

where $*$ denotes the convolution operation. However, due to the restricted size of localized convolutional kernels, CNN-based methods face fundamental mathematical limitations in efficiently capturing long-range spatial dependencies across the entire scene.

B. SSMs for Sequence Modeling

SSMs provide an efficient mathematical framework for sequence modeling. Fundamentally, an SSM maps a 1-D input sequence $\mathbf{x}_t$ to a response $\mathbf{y} * t$ through a continuous-time linear state-space formulation

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t), \qquad \mathbf{y}(t) = \mathbf{C}\mathbf{h}(t) \tag{3}$$

where $\mathbf{h}(t)$ denotes the hidden state, and $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ represent the continuous-time state transition, input, and output matrices, respectively. To apply the continuous-time system to discrete input sequences, the zero-order hold (ZOH) discretization is adopted, assuming that the input remains constant within each sampling interval Δ. The state at the next time step can then be obtained as

$$\mathbf{h}_t = e^{\Delta\mathbf{A}}\mathbf{h}_{t-1} + \left(\int_0^{\Delta} e^{(\Delta-\tau)\mathbf{A}} d\tau\right)\mathbf{B}\mathbf{x}_t. \tag{4}$$

Accordingly, the continuous-time SSM is transformed into the discrete-time formulation

$$\mathbf{h} * t = \bar{\mathbf{A}}\mathbf{h} * t - 1 + \bar{\mathbf{B}}\mathbf{x}_t, \qquad \mathbf{y}_t = \mathbf{C}\mathbf{h} * t \tag{5}$$

where the discretized transition matrix $\bar{\mathbf{A}}$ and input matrix $\bar{\mathbf{B}}$ are given by

$$\bar{\mathbf{A}} = e^{\Delta\mathbf{A}} \tag{6}$$

and

$$\bar{\mathbf{B}} = \left(\int_0^{\Delta} e^{(\Delta-\tau)\mathbf{A}} d\tau\right)\mathbf{B}. \tag{7}$$

When $\mathbf{A}$ is invertible, $\bar{\mathbf{B}}$ can be further written as

$$\bar{\mathbf{B}} = \mathbf{A}^{-1}\left(e^{\Delta\mathbf{A}} - \mathbf{I}\right)\mathbf{B}. \tag{8}$$

This discretization converts the continuous-time dynamics into a recursive form suitable for sequential computation and

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provides the mathematical basis for efficient long-sequence modeling in SSMs.

Nevertheless, applying canonical 1-D Mamba architectures to 2-D HSIs poses a significant structural challenge [30]. Standard raster scanning directly flattens the spatial dimensions row by row, introducing massive spatial discontinuities at row boundaries. This serialization severely disrupts the native 2-D topological structures and local neighborhood contiguity, which are crucial priors for accurate abundance estimation.

C. Frequency-Domain Neural Operators

Frequency-domain analysis, such as the wavelet Transform, has been utilized to decouple low-frequency background structures from high-frequency noise [43], [44]. Recently, FNOs have emerged as a powerful paradigm for continuous global frequency modeling. Given an input feature representation $\mathcal{Z}^{(l)}$, the standard FNO layer operates via the fast Fourier transform (FFT)

$$\mathcal{Z}^{(l+1)} = \sigma\left(W\mathcal{Z}^{(l)} + \mathcal{F}^{-1}\left(R \cdot \mathcal{F}\left(\mathcal{Z}^{(l)}\right)\right)\right) \tag{9}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the FFT and inverse FFT, respectively, and R is a parameterized complex weight tensor. A comparison of the computational complexity and receptive fields of CNN, standard ViT, Swin Transformer, and the target architecture considered in this work is presented in Fig. 1.

Recent studies have also explored FNO-based learning in remote sensing and geophysical applications; for example, Wang et al. [45] developed a transfer learning FNO for frequency-domain wave-equation modeling, demonstrating the potential of FNOs for efficient global physical-field modeling in geophysical exploration. However, the standard FNO relies on a static, globally shared weight tensor R that treats all spatial locations and frequency modes uniformly once trained. High-resolution HSIs inevitably suffer from spatially varying environmental noise and spectral variability. The rigid filtering mechanism of standard FNOs lacks the dynamic weighting capacity required to adaptively recalibrate low-frequency modes and suppress complex high-frequency disturbances. This limitation necessitates a more flexible multistream paradigm capable of dynamically reweighing global frequency representations while synergizing with spatial models.

III. Proposed Method

To effectively address the inherent limitations of local receptive fields in CNNs [20] and the quadratic computational complexity of Transformers, SSMs [30] have recently emerged as a highly promising paradigm for modeling long-range dependencies. However, directly applying canonical 1-D Mamba architectures to 2-D HSI inherently disrupts local spatial topological continuity. Furthermore, pure spatial sequential scanning lacks a global perspective in the frequency domain, making the unmixing process highly susceptible to high-frequency environmental noise and complex spectral variability.

Motivated by these challenges, we propose a novel dual-stream spatial-frequency SSM, termed WF-Mamba, tailored for HU. The core philosophy of WF-Mamba lies in the elegant parallel integration of microscopic spatial sequence inference (enabled by physics-driven SS) and macroscopic global frequency purification (WFNO [42]). By embedding this dual-stream mechanism within a multifrequency wavelet AE framework, the proposed method comprehensively captures both local continuous structures and global frequency consistency.

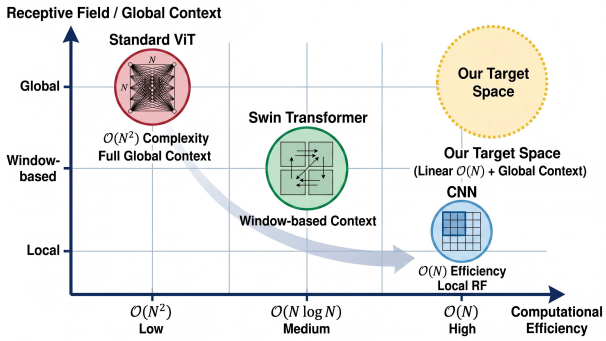


Fig. 1. Comparison of computational complexity and receptive field among CNN, standard ViT, Swin Transformer, and our target space.

The detailed formulation of the proposed WF-Mamba framework is systematically presented in section III. First, Section III-A provides a macroscopic overview of the multistream wavelet AE architecture. Subsequently, the core architectural innovation—the visual Mamba layer equipped with parallel spatial/frequency branches and an adaptive synergistic fusion (ASF) mechanism—is comprehensively detailed in Section III-B. Finally, the physics-constrained objective functions for network optimization are formulated in Section III-C.

A. Overall Architecture of WF-Mamba

The HU task fundamentally relies on the LMM formulated in (1). To robustly estimate the endmember matrix $\mathbf{M}$ and the corresponding fractional abundances $\mathbf{A}$ under complex environmental noise, the proposed WF-Mamba is constructed as an unsupervised multistream AE. The overall architecture is illustrated in Fig. 2.

To explicitly decouple intrinsic material structures from high-frequency environmental interference, the network abandons the conventional single-input paradigm. Instead, it employs a 2-D stationary wavelet transform (2-D SWT) to decompose the input HSIs $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$ into four distinct multifrequency streams

$$\{\mathbf{X}_A, \mathbf{X}_H, \mathbf{X}_V, \mathbf{X}_D\} = \mathcal{W}_{SWT}(\mathbf{X}). \tag{10}$$

Here, $\mathbf{X}_A$ denotes the low-frequency approximation component, which mainly preserves the large-scale spatial structures, background information, and relatively smooth material distributions of the HSI. $\mathbf{X}_H$, $\mathbf{X}_V$, and $\mathbf{X}_D$ represent the horizontal, vertical, and diagonal high-frequency detail components, respectively, which primarily characterize directional spatial variations, object boundaries, fine textures, and local structural discontinuities. In the context of HU, $\mathbf{X}_A$ provides relatively stable global material information, whereas the three detail streams preserve complementary local structural cues. This multifrequency decomposition enables the proposed

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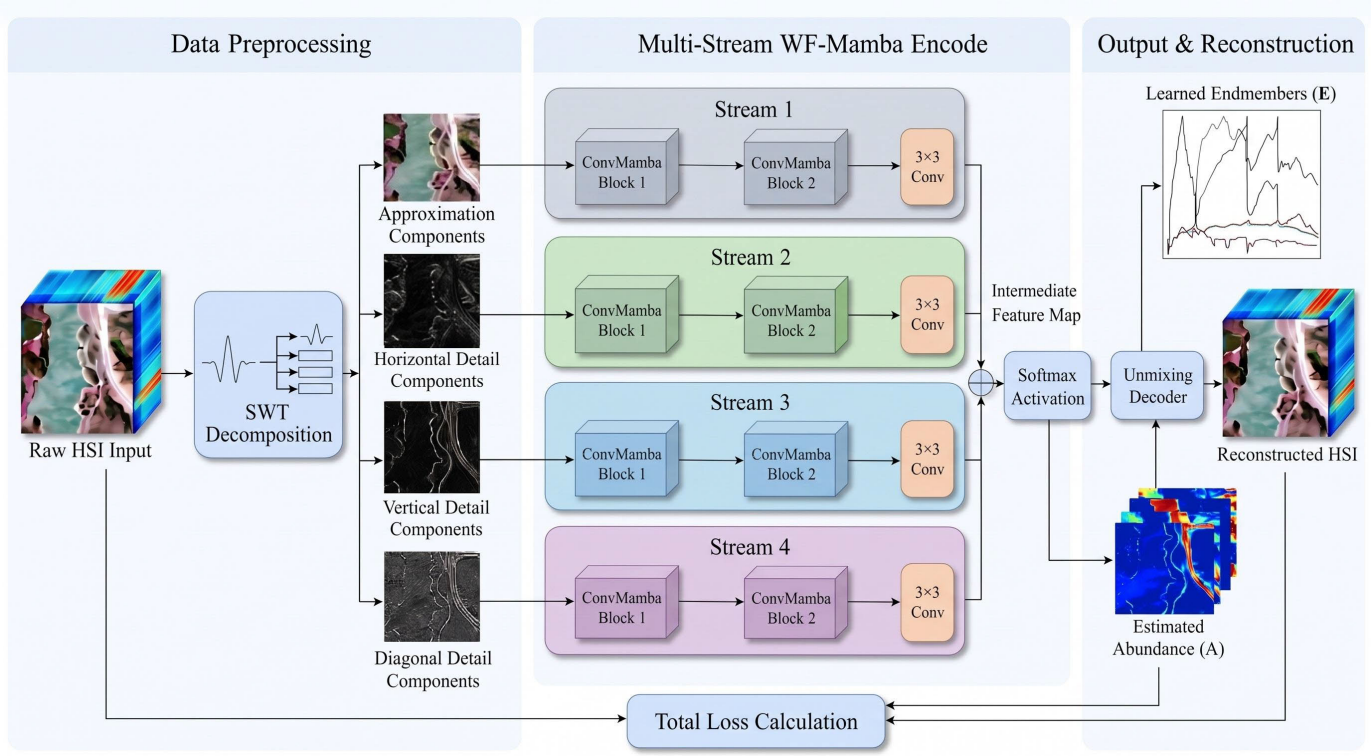


Fig. 2. Overall multistream architecture of the proposed WF-Mamba framework. The input HSI is decomposed by 2-D SWT into four streams, which are subsequently processed by parallel spatial-frequency encoders and reconstructed via a physically constrained linear decoder.

network to jointly exploit large-scale spatial consistency and fine-grained boundary information while improving robustness to noise.

The encoder processes these four streams in parallel. Each stream is independently processed by two cascaded ConvMambaBlocks, followed by a 3×3 convolutional layer for feature projection. The outputs from the four parallel branches are then aggregated via elementwise summation to obtain the fused latent representation $\mathbf{H}$.

To convert $\mathbf{H}$ into physically meaningful fractional abundances, an abundance head with a softmax activation function is employed. For the jth pixel, the estimated abundance of the ith endmember $\mathbf{A}_{i,j}$ is

$$\mathbf{A}_{i,j} = \frac{\exp\left(\mathbf{H}_{i,j}\right)}{\sum_{r=1}^{R} \exp\left(\mathbf{H}_{r,j}\right)}. \tag{11}$$

This operation naturally and strictly enforces the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC) required by physical realities.

Finally, the decoder is implemented as a linear 1×1 convolutional layer without a bias term. Let its weight tensor be denoted by $\mathbf{W}_D \in \mathbb{R}^{L \times R \times 1 \times 1}$, where L and R represent the number of spectral bands and endmembers, respectively. For the jth pixel, the reconstruction operation can be written as

$$\hat{\mathbf{x}}_j = \mathbf{W}_D \mathbf{a}_j. \tag{12}$$

Since the kernel size is 1×1 and no bias term is introduced, the decoder weights directly correspond to the endmember matrix after removing the two singleton spatial dimensions

$$\mathbf{M} = \text{squeeze}\left(\mathbf{W}_D\right) \in \mathbb{R}^{L \times R}. \tag{13}$$

Therefore, the network reconstructs the HSI according to $\hat{\mathbf{X}} = \mathbf{M} * \mathbf{A}$, where each column of $\mathbf{M}$ represents an endmember spectrum and each column of $\mathbf{A}$ contains the corresponding abundance fractions. This direct correspondence between the decoder parameters and the endmember matrix provides an explicit physical interpretation of the reconstruction process.

B. Visual Mamba Layer With Spatial-Frequency Parallelism

The core architectural engine of WF-Mamba is the *Visual Mamba Layer*, which transcends the limitations of standard 1-D SSMs in processing 2-D hyperspectral data. As illustrated in Fig. 3, this layer is fundamentally composed of three synergistic components: spatial position anchoring, a microscopic spatial Mamba branch, and a macroscopic frequency WFNO branch, culminating in an adaptive fusion module.

To prevent the permanent loss of absolute coordinate priors during 1-D serialization, the layer first injects a learnable 2-D spatial positional embedding $\mathbf{P} \in \mathbb{R}^{C \times H_p \times W_p}$. For an input feature map $\mathbf{X}$, the position-aware feature is generated as $\tilde{\mathbf{X}} = \mathbf{X} + \mathcal{F}_{\text{interp}}(\mathbf{P}, (H, W))$, where $\mathcal{F}_{\text{interp}}$ denotes bilinear interpolation.

1) Microscopic Spatial Branch via SS: Standard raster scanning introduces massive spatial discontinuities at row boundaries, where physically adjacent pixels become distant in the 1-D sequence. To maintain structural contiguity, we utilize a *SS* mechanism.

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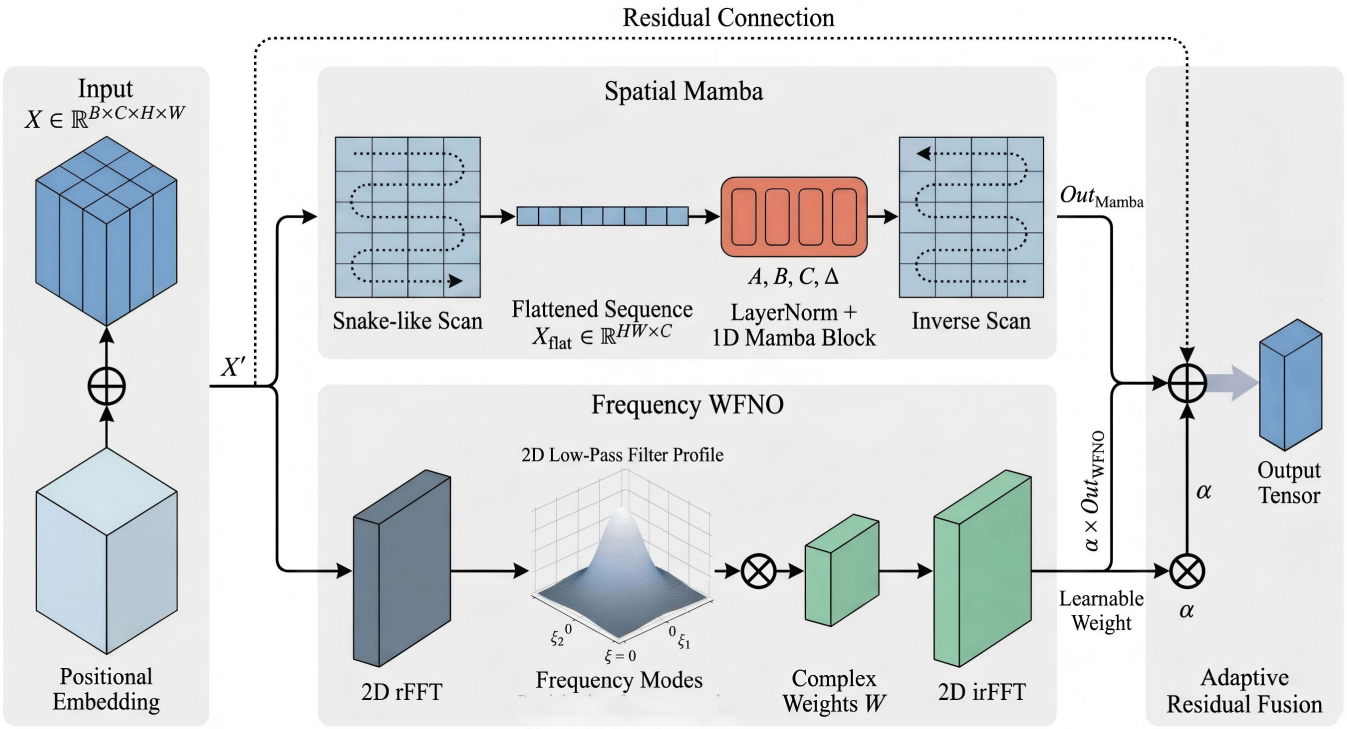


Fig. 3. Detailed structure of the visual Mamba layer, illustrating the parallel integration of the microscopic spatial Mamba branch (via SS), the macroscopic frequency WFNO branch, and the adaptive residual fusion mechanism.

For a pixel at spatial coordinates (i, j) in $\tilde{\mathbf{X}}$ $(0 \le i < H, 0 \le j < W)$, its position k in the 1-D flattened sequence $\mathbf{S} \in \mathbb{R}^{HW\times C}$ is formulated by the mapping

$$k = i \times W + \begin{cases} j, & \text{if } i \text{ is even} \\ W - 1 - j, & \text{if } i \text{ is odd} \end{cases} \tag{14}$$

This alternating trajectory preserves local neighborhood priors. Compared with conventional raster scanning, the snake-like trajectory avoids abrupt directional changes and spatial discontinuities at row boundaries, where physically adjacent pixels may otherwise become distant or weakly connected in the serialized sequence. By maintaining a more spatially coherent ordering, neighboring pixels with strong spatial correlation or local structure can be modeled consecutively by the Mamba block, which facilitates the propagation of local spatial information and preserves the continuity of abundance structures. The sequence $\mathbf{S}$ is then processed by the Mamba block, which implements a continuous-time SSM discretized via the ZOH

$$\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}\mathbf{s}_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t \tag{15}$$

where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ represent the discretized transition parameters. The sequential output is transformed back to the 2-D spatial domain via an inverse snake-like scan, yielding the spatial topological feature $\mathbf{X}_{Mamba}$.

2) Macroscopic Frequency Branch via WFNO: While the spatial Mamba branch excels at sequential modeling, it is susceptible to localized noise perturbations. To achieve global spectral–spatial consistency, we design a parallel WFNO branch. The frequency-domain processing procedure of the WFNO branch is illustrated in Fig. 4. The feature $\tilde{\mathbf{X}}$ is first projected into the frequency domain via a 2-D real FFT (rFFT)

$$\mathbf{Z} = \mathcal{F}_{rFFT2}\left(\tilde{\mathbf{X}}\right) \in \mathbb{C}^{C\times H\times\left(\frac{W}{2}+1\right)} \tag{16}$$

To achieve frequency-domain noise filtering, we apply a learnable complex weight tensor $\mathbf{W} \in \mathbb{C}^{C\times C\times M_1\times M_2}$, where M_1 and M_2 denote the numbers of retained low-frequency modes along the two spatial dimensions, respectively. Specifically, only the low-frequency region of $\mathbf{Z}$ corresponding to the first M_1 and M_2 frequency modes is retained for subsequent processing, while the remaining high-frequency components are discarded. The truncated frequency representation can be expressed as

$$Z^{\text{LF}}_{b,c,x,y} = Z_{b,c,x,y}\,\mathbb{I}\left(x < M_1,\ y < M_2\right). \tag{17}$$

The rebalancing operation is then executed via complex Einstein summation

$$\hat{\mathbf{Z}}_{b,o,x,y} = \sum_{i=1}^{C} \mathbf{W}_{i,o,x,y} \cdot \mathbf{Z}^{\text{LF}}_{b,i,x,y} \tag{18}$$

where $x = 0, \ldots, M_1 - 1$ and $y = 0, \ldots, M_2 - 1$ denote the retained frequency indices. The processed low-frequency representation is embedded into the original Fourier domain, with the discarded high-frequency components set to zero. Finally, the filtered frequency representation is mapped back to the spatial domain via the inverse real-valued Fourier transform

$$\mathbf{X}_{WFNO} = \mathcal{F}^{-1}_{rFFT2}\left(\hat{\mathbf{Z}}\right) \tag{19}$$

yielding a globally filtered spatial feature representation. By retaining and adaptively reweighting only the low-frequency

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TABLE I
SENSITIVITY ANALYSIS OF THE INITIAL FUSION WEIGHT α ON DIFFERENT DATASETS

	Houston		Moffett		JS	
	RMSE	SAD	RMSE	SAD	RMSE	SAD
0.3	0.081	0.075	0.053	0.019	0.090	0.054
0.4	0.082	0.077	0.054	0.022	0.084	0.055
0.5	**0.075**	**0.074**	**0.053**	**0.017**	**0.083**	**0.052**
0.6	0.080	0.076	0.053	0.021	0.088	0.054
0.7	0.078	0.077	0.053	0.023	0.085	0.054

modes, the WFNO branch acts as a dynamic physical low-pass filter to suppress high-frequency noise while preserving the dominant global spatial structures.

3) Noise-Robust ASF: The distinct characteristics of the Mamba and WFNO branches necessitate a dynamic integration strategy rather than a rigid concatenation. Since the multiple wavelet streams demand different levels of spatial precision and frequency smoothing, we propose a learnable adaptive residual fusion mechanism

$$\mathbf{X}_{out} = \tilde{\mathbf{X}} + \mathbf{X}_{Mamba} + \alpha \cdot \mathbf{X}_{WFNO} \tag{20}$$

where $\alpha \in \mathbb{R}$ is a learnable gating parameter (initialized to 0.5) that dictates the contribution of the frequency domain.

To investigate the sensitivity of the fusion weight initialization, we further evaluate the unmixing performance with different initial values of α ranging from 0.3 to 0.7. Note that α is optimized during training, and this sensitivity analysis mainly evaluates the influence of different initializations rather than fixing α values. The results are reported in Table I. It can be observed that the model maintains relatively stable performance over the tested range, indicating that the proposed ASF mechanism is not overly sensitive to the initialization of α. Among the tested settings, $\alpha = 0.5$ provides the most balanced overall performance across the three datasets, achieving the best results on both root-mean-square error (RMSE) and spectral angle distance (SAD) for Houston and Jasper Ridge, while obtaining the lowest SAD on Moffett. Therefore, $\alpha = 0.5$ is adopted as the default initialization in the following experiments.

During training, the network autonomously learns to balance the crisp local spatial boundaries extracted by $\mathbf{X}_{Mamba}$ and the global noise-purified representations provided by $\mathbf{X}_{WFNO}$, ensuring optimal synergistic unmixing across all frequency bands.

C. Objective Function and Optimization

HU is inherently a highly illposed inverse problem. Relying solely on the unsupervised reconstruction error is insufficient to obtain physically meaningful abundance maps. Therefore, we design a physics-driven objective function that considers spectral fidelity, microscopic sparsity, and macroscopic spatial smoothness.

1) SAD Loss: To ensure robust spectral fidelity against illumination variations and topographical undulations, we adopt the SAD loss

$$\mathcal{L}_{SAD} = \frac{1}{N}\sum_{j=1}^{N} \arccos\left(\frac{\mathbf{x}_j^\top \hat{\mathbf{x}}_j}{\|\mathbf{x}_j\|_2 \, \|\hat{\mathbf{x}}_j\|_2}\right) \tag{21}$$

where N denotes the total number of pixels. SAD is invariant to linear amplitude scaling, preserving the intrinsic spectral characteristics.

2) Abundance Sparsity Constraint: Given that a local mixed pixel is typically composed of only a few endmembers, we impose an L_1-norm regularization on the estimated abundance tensor $\mathbf{A}$ to suppress redundant activations

$$\mathcal{L}_{L_1} = \frac{1}{N}\sum_{j=1}^{N} \|\mathbf{a}_j\|_1 . \tag{22}$$

This microscopic sparsity prior serves as an Occam's razor to yield cleaner abundance maps.

3) TV Regularization: To enforce macroscopic piecewise spatial smoothness and alleviate isolated mottled artifacts, we introduce the anisotropic total variation (TV) penalty

$$\mathcal{L}_{TV} = \sum_{i=1}^{R}\sum_{h,w}\left(\left|\mathbf{A}_{i,h+1,w} - \mathbf{A}_{i,h,w}\right| + \left|\mathbf{A}_{i,h,w+1} - \mathbf{A}_{i,h,w}\right|\right). \tag{23}$$

By penalizing abrupt changes between adjacent pixels, the TV constraint ensures spatially consistent abundance distributions.

4) Joint Optimization Strategy: The overall objective function is formulated as a weighted sum of the aforementioned physical constraints

$$\mathcal{L}_{total} = \mathcal{L}_{SAD} + \lambda_{L_1}\mathcal{L}_{L_1} + \lambda_{TV}\mathcal{L}_{TV} \tag{24}$$

where λ_{L_1} and λ_{TV} are hyperparameters dynamically adapted to account for the varying physical complexities across different datasets. The entire framework is end-to-end optimized using the Adam optimizer with a learning rate scheduler.

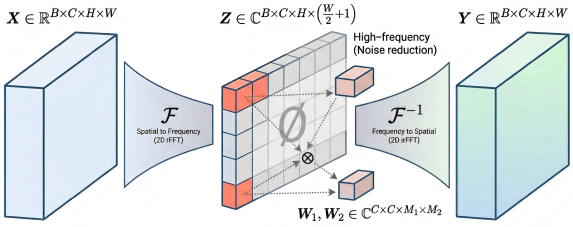
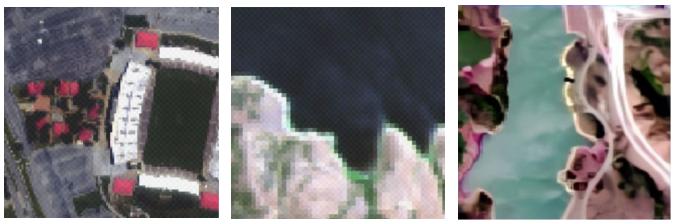


Fig. 4. Illustration of the WFNO operating in the frequency domain. It dynamically recalibrates low-frequency modes while filtering out high-frequency noise representations.

IV. EXPERIMENTS

To comprehensively evaluate the performance of the proposed WF-Mamba, we conduct experiments on one synthetic dataset and three widely used real-world hyperspectral datasets (their pseudocolor images are visualized in Fig. 5): Houston, Moffett field, and Jasper Ridge.

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(a) (b) (c)

Fig. 5. Pseudocolor images of the three real-world hyperspectral datasets used in the experiments. (a) Houston. (b) Moffett field. (c) Jasper Ridge.

A. Datasets

1) Synthetic Dataset: The synthetic dataset is generated to provide a controlled environment for performance validation under known ground truth (GT). We select $R = 5$ endmembers from the USGS spectral library to simulate the mixing process. The dataset consists of 130 $\times$ 130 pixels with 166 spectral bands. The fractional abundances are generated following a Dirichlet distribution to satisfy the ANC and ASC constraints.

2) Jasper Ridge: The Jasper Ridge dataset is a popular benchmark in HU, captured by the AVIRIS sensor. In our experiments, a sub-image of 100 $\times$ 100 pixels is utilized. After removing the water absorption and low-SNR bands, 198 spectral bands are retained. This scene primarily contains four typical endmembers: road, soil, water, and tree.

3) Houston: The Houston dataset was originally provided by the 2013 IEEE GRSS Data Fusion Contest. For the unmixing task, we selected a representative region characterized by complex urban structures. The dataset features a high spatial resolution and contains diverse endmembers such as grass, soil, and various man-made materials. Due to the presence of intricate shadow interference and diverse backgrounds, this dataset poses a significant challenge for stable abundance estimation, requiring strong sparsity and spatial regularization.

4) Moffett Field: This dataset was also acquired by the AVIRIS sensor over the Moffett field area in California. A subscene of 50 $\times$ 50 pixels is extracted for validation. After preprocessing to eliminate noisy bands, 189 bands are utilized for the unmixing process. The scene mainly consists of three ground objects: Water, Soil, and Vegetation.

B. Evaluation Metrics

To quantitatively assess the performance of the proposed WF-Mamba and the benchmark methods, we utilize two widely recognized metrics in the HU community: SAD and RMSE.

1) Spectral Angle Distance: The SAD is a crucial index to evaluate the similarity between the estimated endmember spectral signatures $\hat{\mathbf{e}}_i$ and the reference GT $\mathbf{e}_i$. It is defined as the multidimensional angle between two vectors

$$\mathrm{SAD}_i = \arccos\left(\frac{\mathbf{e}_i^\top \hat{\mathbf{e}}_i}{\|\mathbf{e}_i\|_2 \, \|\hat{\mathbf{e}}_i\|_2}\right). \tag{25}$$

A smaller SAD value signifies that the estimated endmember curve more closely matches the physical signature of the target material. Notably, SAD is invariant to linear amplitude scaling, making it a robust metric under varying illumination and shadow conditions. We report the SAD for each endmember along with the mean SAD (mSAD) across all endmembers.

2) Root-Mean-Square Error: The RMSE is employed to evaluate the accuracy of the estimated fractional abundance maps at the pixel level. For the ith endmember, the RMSE is formulated as

$$\mathrm{RMSE}_i = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\left(a_{i,j} - \hat{a}_{i,j}\right)^2} \tag{26}$$

where $a_{i,j}$ and $\hat{a}_{i,j}$ represent the reference and estimated abundance fractions for the jth pixel, respectively, and N is the total number of pixels. A lower RMSE implies a more precise estimation of material distributions.

C. Implementation Details

All experiments are conducted on a unified hardware and software platform to ensure fair and reproducible comparisons. The hardware system is equipped with an Intel(R) Xeon(R) CPU E5-2696 v4 @ 2.20 GHz, 128 GB of RAM, and four NVIDIA Tesla V100-SXM2-16GB GPUs. The software environment is built upon the Ubuntu 24.04 operating system with CUDA version 12.2, and the proposed WF-Mamba framework is implemented using the PyTorch deep learning library.

Since the proposed method formulates HU as an unsupervised deep AE task, the target HSI is directly fed into the network without spatial patch cropping or training-test subset splitting (i.e., the batch size is strictly set to 1). To mitigate the inherent randomness introduced by network initialization, all experiments are independently executed 10 times, and the average quantitative metrics are reported.

The entire network is optimized end-to-end utilizing the Adam optimizer. To guarantee stable convergence and fine-tune the parameters in the later stages of training, a step-based learning rate scheduler (StepLR) is adopted to dynamically decay the learning rate by a specific factor (γ) at given step intervals. The parameters of the 2-D spatial positional embeddings and the WFNO complex weights are initialized dynamically according to the varying spatial dimensions of the input wavelets.

As detailed in Table II, the crucial hyperparameter configurations are adaptively tailored to the specific physical complexities of each dataset. For instance, the Houston dataset captures a highly complex urban scene with severe background noise and varied structures. Consequently, we explicitly increase the L_1 sparsity penalty weight ($\lambda_{L_1} = 5 \times 10^{-3}$) to effectively suppress noise interference and enforce a more rigid sparsity prior. In contrast, for relatively pure natural scenes like Jasper Ridge, milder regularization terms ($\lambda_{L_1} = \lambda_{TV} = 1 \times 10^{-4}$) are sufficient to yield optimal unmixing results.

D. Comparison With Advanced Methods

To comprehensively evaluate the superiority and robustness of the proposed WF-Mamba framework, we conduct comparative experiments on the synthetic dataset and three widely used real-world hyperspectral datasets: Houston, Jasper Ridge,

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TABLE II
DETAILED HYPERPARAMETER CONFIGURATIONS FOR DIFFERENT DATASETS

Dataset	Epochs	Dropout	Learning Rate	Step Size	γ	Weight Decay	λ_{L_1}	λ_{TV}
Synthetic	300	0.0	1×10^{-3}	50	0.8	1×10^{-4}	1×10^{-3}	1×10^{-3}
Jasper Ridge	250	0.1	9×10^{-4}	50	0.8	1×10^{-3}	1×10^{-4}	1×10^{-4}
Houston	260	0.1	5×10^{-3}	40	0.8	1×10^{-3}	5×10^{-3}	1×10^{-4}
Moffett Field	280	0.1	14×10^{-3}	35	0.5	5×10^{-4}	1×10^{-4}	1×10^{-4}

TABLE III
QUANTITATIVE RESULTS ON THE SYNTHETIC DATASET WITH DIFFERENT SNRS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST ARE UNDERLINED

Metric	SNR	DFFN	A2SN	UNMamba	CyCUNet	SWC	SSAF	DTU	WF-Mamba
SAD	10dB	0.149	0.185	0.137	0.166	0.125	0.187	0.171	**0.119**
	20dB	0.116	0.142	0.100	0.130	0.101	0.147	0.122	**0.082**
	30dB	0.088	0.115	0.084	0.098	0.079	0.116	0.103	**0.067**
RMSE	10dB	0.185	0.213	0.162	0.195	0.156	0.220	0.201	**0.132**
	20dB	0.141	0.179	0.140	0.158	0.123	0.176	0.152	**0.104**
	30dB	0.111	0.145	0.100	0.129	0.093	0.142	0.113	**0.082**

TABLE IV
PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE HOUSTON DATASET. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND-BEST RESULTS ARE UNDERLINED

Method		DFFN	A2SN	UNMamba	CyCUNet	SWC	SSAF	DTU	WF-Mamba
SAD	Grass	0.092	0.076	0.073	0.099	0.058	0.114	0.091	**0.053**
	Runway	0.066	0.284	0.101	0.113	0.099	0.088	**0.064**	0.103
	Parking lot1	0.065	0.173	0.061	0.072	**0.057**	0.257	0.167	0.059
	Parking lot2	**0.079**	0.346	0.184	0.145	0.091	0.218	0.256	0.082
	Average	0.076	0.220	0.105	0.107	0.077	0.169	0.145	**0.074**
RMSE	Grass	0.070	0.054	0.068	0.094	0.045	0.070	0.070	**0.042**
	Runway	0.076	0.270	0.059	0.181	0.042	0.082	0.040	**0.039**
	Parking lot1	0.283	0.161	0.140	0.325	0.115	0.262	**0.095**	0.108
	Parking lot2	0.253	0.218	0.106	0.223	**0.105**	0.257	**0.105**	0.109
	Average	0.171	0.176	0.099	0.206	0.076	0.168	0.077	**0.075**

and Moffett field. Seven advanced (SOTA) HU methods are selected as benchmarks, covering representative deep convolutional architectures and attention-based models: DFFN [46], A2SN [22], UNMamba [30], CyCUNet [47], SWC [36], SSAF [48], and DTU [28]. A thorough analysis is performed from two dimensions: quantitative evaluation metrics (SAD and RMSE) and qualitative visualization (abundance maps and extracted endmember signatures).

1) Results on Synthetic Dataset: To evaluate the model's robustness against varying degrees of environmental interference, we first conduct experiments on the synthetic dataset under three different noise levels: SNR = 10, 20, and 30 dB. The quantitative results of SAD and RMSE are summarized in Table III.

As observed in Table III, WF-Mamba exhibits a clear and consistent performance improvement as the noise level decreases. Specifically, as the SNR increases from 10 to 30 dB, the SAD decreases from 0.119 to 0.082 and further to 0.067, while the RMSE decreases from 0.132 to 0.104 and further to 0.082. This consistent downward trend indicates that the unmixing accuracy of WF-Mamba improves steadily as the noise level decreases. Notably, even under the severe noise condition of 10 dB, our method achieves the lowest SAD and RMSE among all compared methods, demonstrating strong robustness to heavily contaminated observations. This robustness is mainly attributed to the complementary roles of the SWT-based multistream representation and the WFNO branch, where the former captures multifrequency spatial structures while the latter dynamically recalibrates low-frequency modes and suppresses high-frequency noise disturbances in the frequency domain.

2) Quantitative Performance Comparison on Real-World Datasets: Real-world HSIs inherently contain complex nonlinear mixing effects, intraclass spectral variability, and spatial heterogeneity, which pose significantly greater challenges than synthetic datasets. To comprehensively evaluate the

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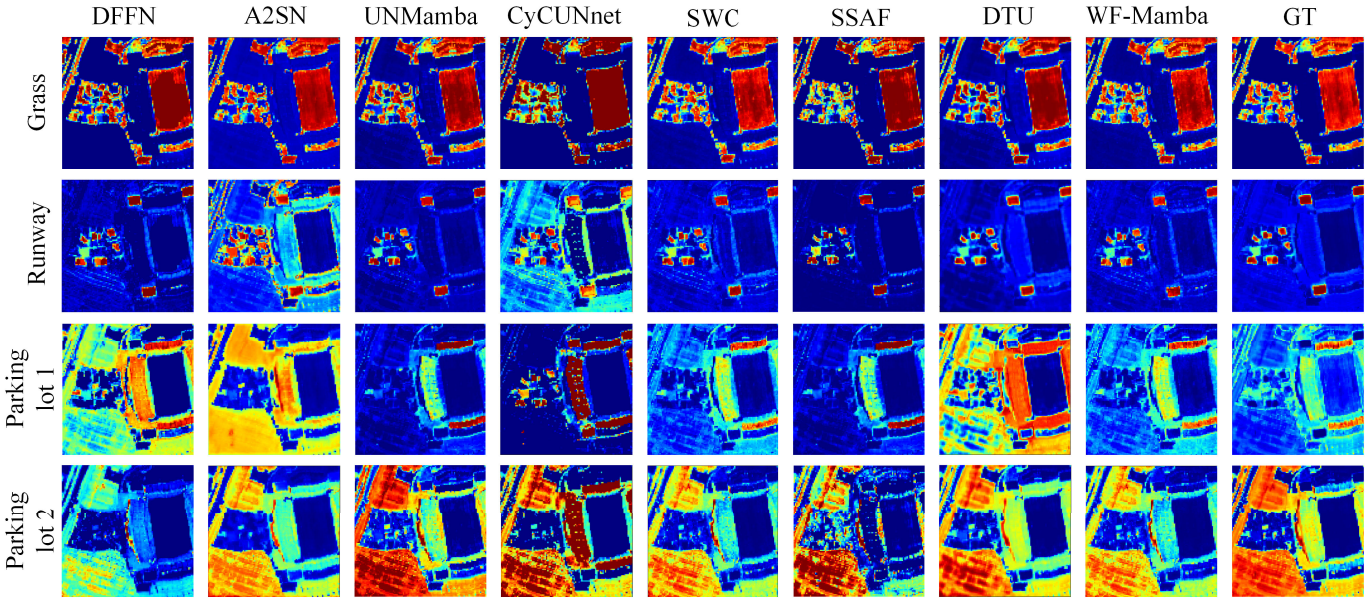


Fig. 6. Visualization of resulting abundance and GT on Houston.

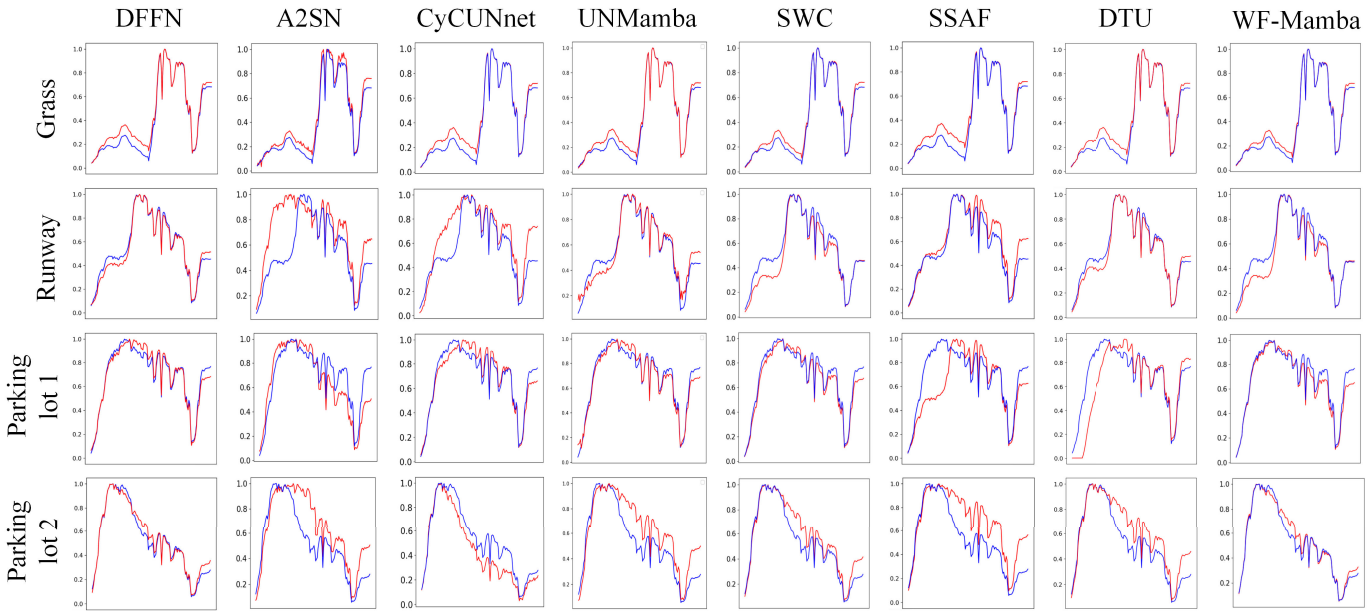


Fig. 7. Comparison of extracted endmember signatures on the Houston dataset. The red lines represent the GT, while the blue lines denote the estimated signatures by different methods.

generalization ability and practical applicability of the proposed WF-Mamba framework, we conduct extensive experiments on three benchmark datasets: Houston, Moffett field, and Jasper Ridge. The quantitative performance comparison results on the three real-world datasets are summarized in Table IV for Houston, Table V for Moffett Field, and Table VI for Jasper Ridge, respectively. These results provide a comprehensive quantitative assessment of the unmixing accuracy of WF-Mamba in terms of SAD and RMSE under different scene characteristics.

a) Houston dataset (complex urban scene): From the abundance maps in Fig. 6, WF-Mamba produces cleaner and more spatially continuous distributions than the competing methods. In particular, for the grass component, the boundary between the grass-covered region and the surrounding background is reconstructed more continuously, with fewer isolated noisy responses near the object edges. For the runway component, the proposed method better preserves the elongated structure and its boundary, while avoiding the fragmented responses observed in several competing methods. For the two parking lot components, WF-Mamba also provides more coherent responses within the large target regions and reduces irregular artifacts around their boundaries.

The corresponding endmember curves in Fig. 7 further show that WF-Mamba maintains a close agreement with the reference signatures around the dominant spectral peaks and valleys. Although some fine high-frequency fluctuations are slightly smoothed, the overall spectral shape is better pre-

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TABLE V
PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE MOFFETT DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method		DFFN	A2SN	UNMamba	CyCUNet	SWC	SSAF	DTU	WF-Mamba
SAD	Water	0.029	0.043	0.044	<u>0.020</u>	0.021	0.141	<u>0.020</u>	**0.016**
	Soil	0.014	0.099	0.152	0.014	<u>0.011</u>	0.022	0.025	**0.009**
	Vegetation	0.080	0.282	0.177	<u>0.018</u>	**0.017**	0.045	0.112	0.024
	Average	0.039	0.141	0.124	<u>0.017</u>	<u>0.017</u>	0.069	0.052	**0.016**
RMSE	Water	0.130	0.121	0.106	0.120	0.075	**0.024**	0.068	<u>0.054</u>
	Soil	0.076	0.094	0.171	0.059	0.060	**0.037**	0.066	<u>0.055</u>
	Vegetation	0.110	0.144	0.209	0.079	0.053	<u>0.052</u>	<u>0.052</u>	**0.045**
	Average	0.105	0.119	0.167	0.086	0.063	**0.038**	0.062	<u>0.051</u>

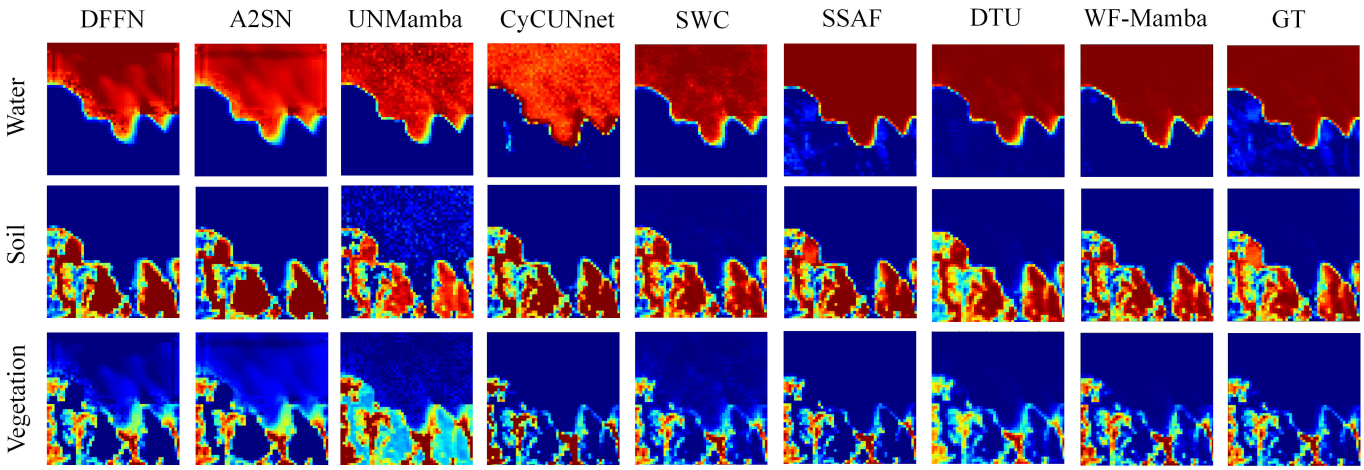


Fig. 8. Visualization of resulting abundance and GT on Moffett.

served, indicating that the proposed frequency-domain filtering suppresses noise while retaining the major material-dependent spectral characteristics.

b) Moffett field dataset (coastal zone scene): As shown in Fig. 8, WF-Mamba provides a smoother and more coherent representation of the Water component, especially along the irregular land-water transition boundary, where several competing methods exhibit noisy or discontinuous responses. For the Soil component, the proposed method preserves the main spatial structures while suppressing scattered responses in the background. In the vegetation map, WF-Mamba better maintains the continuity of vegetation regions and the transitions between vegetation and surrounding materials, although some very fine local variations are slightly smoothed.

In Fig. 9, the estimated water and soil signatures closely follow the reference curves over the dominant spectral regions, while the Vegetation signature remains consistent around its major spectral peaks and valleys. These results indicate that the proposed method improves spatial coherence without causing severe distortion of the dominant spectral characteristics.

c) Jasper ridge dataset (natural mixed scene): The abundance maps in Fig. 10 show that WF-Mamba produces smoother and more spatially coherent distributions for the water and road components, with clearer boundaries between the target regions and their surroundings. This is particularly visible along the narrow linear structures of the road component, where competing methods tend to generate fragmented or noisy responses. For the Tree and Grass components, WF-Mamba suppresses isolated artifacts while preserving the major spatial regions, resulting in abundance maps that are closer to the reference distributions.

As shown in Fig. 11, WF-Mamba also provides stable spectral reconstruction for grass, tree, and roof, with the estimated curves closely following the reference signatures around the dominant peaks, valleys, and major local variations. For Asphalt, the estimated curve is slightly smoother in some high-frequency regions, which is consistent with the observed tradeoff between fine edge preservation and global noise suppression.

E. Ablation Experiment

To comprehensively evaluate the contribution of the proposed components, we conduct systematic ablation experiments on the Houston, Moffett field, and Jasper Ridge datasets. The quantitative results of the ablation study are summarized in Table VII. In addition to the modulewise ablations of the SS, WFNO, and ASF, we further include the original Mamba configuration with conventional raster scanning as a

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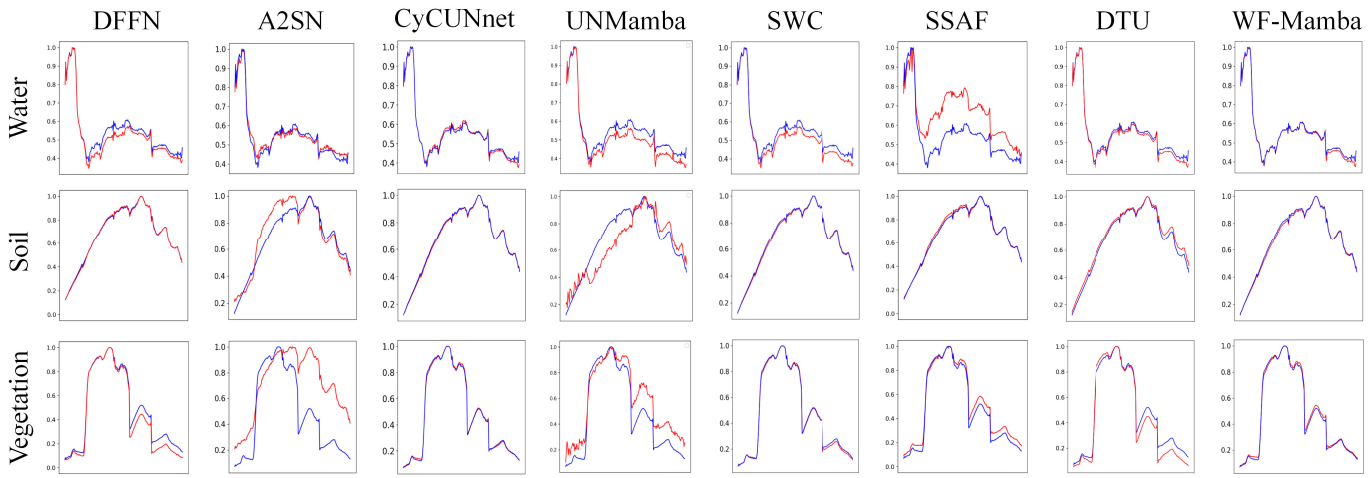


Fig. 9. Comparison of extracted endmember signatures on the Moffett dataset. The red lines represent the GT, while the blue lines denote the estimated signatures by different methods.

TABLE VI

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method		DFFN	A2SN	UNMamba	CyCUNet	SWC	SSAF	DTU	WF-Mamba
SAD	Asphalt	0.135	0.101	0.060	0.121	0.067	**0.037**	0.121	0.065
	Grass	0.045	0.102	0.037	0.082	0.037	0.246	0.058	**0.035**
	Tree	0.113	0.143	0.086	0.101	0.062	**0.021**	0.113	0.058
	Roof	0.091	0.411	0.141	0.052	0.055	**0.023**	0.107	0.049
	Average	0.096	0.196	0.081	0.089	0.054	0.082	0.100	**0.051**
RMSE	Asphalt	0.131	0.136	**0.054**	0.098	0.087	0.089	0.102	0.079
	Grass	0.052	0.182	**0.031**	0.122	0.072	0.111	0.047	0.070
	Tree	0.149	0.191	0.086	0.140	0.111	**0.077**	0.149	0.102
	Roof	0.102	0.242	0.071	0.113	0.095	**0.065**	0.129	0.080
	Average	0.108	0.188	**0.061**	0.118	0.091	0.086	0.107	0.082

baseline. Unless otherwise specified, the network architecture and training settings are kept unchanged for fair comparison.

Frequency-Domain Gain of WFNO: The necessity of the macroscopic frequency-domain branch is robustly confirmed by comparing the "Only SS" configuration, which solely relies on pure spatial-domain processing, with the full model. The experiments demonstrate that introducing the WFNO branch yields a substantial reduction in unmixing errors across all datasets. Taking the Moffett field dataset as an example, the SAD significantly drops from 0.021 to 0.016, and the RMSE decreases from 0.056 to 0.051. This performance leap compellingly proves that pure spatial-domain SSMs struggle to completely eliminate the interference of high-frequency environmental noise, whereas WFNO effectively captures global spectral–spatial consistency by dynamically filtering out high-frequency noise.

Dynamic Balance of ASF: The critical contribution of the ASF mechanism can be assessed by comparing the "SS + WFNO" configuration (which uses simple static addition for branch fusion) with the full model. Experimental results indicate that replacing static addition with ASF consistently and stably enhances unmixing precision. On the Jasper Ridge dataset, the ASF optimization reduces the SAD from 0.055 to 0.051 and the RMSE from 0.090 to 0.082. This thoroughly demonstrates that a rigid equal-weight summation cannot adapt to the varying noise distributions of different wavelet streams. ASF provides the necessary flexibility for the model to dynamically balance crisp spatial boundaries and global frequency smoothness.

Neighborhood Continuity Advantage of SS: To evaluate the effectiveness of our proposed sequence scanning strategy, we compared the configuration using standard raster scanning against the full model. The results reveal that the SS mechanism yields consistent improvements in abundance estimation performance, such as further reducing the SAD from 0.023 to 0.016 and the RMSE from 0.060 to 0.051 on the Moffett field dataset. This validates that preserving the continuity of physically adjacent pixels at row boundaries can prevent abrupt disruptions in local 2-D spatial structures, providing a more spatially coherent sequence for Mamba-based long-range dependency modeling.

F. Network Parameter Sensitivity Analysis

The learning rate is a crucial hyperparameter in deep learning, particularly for architectures integrating frequency-domain neural operators and SSMs, as it directly governs convergence stability and the global feature recalibration process.

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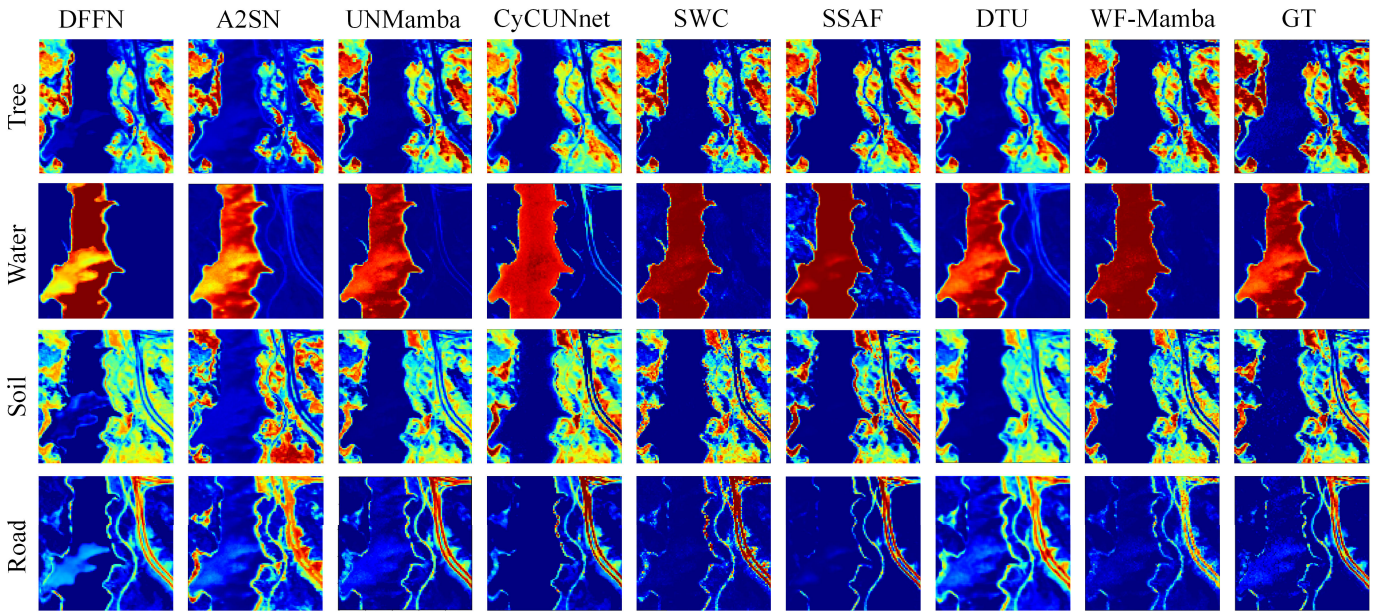


Fig. 10. Visualization of resulting abundance and GT on Jasper Ridge.

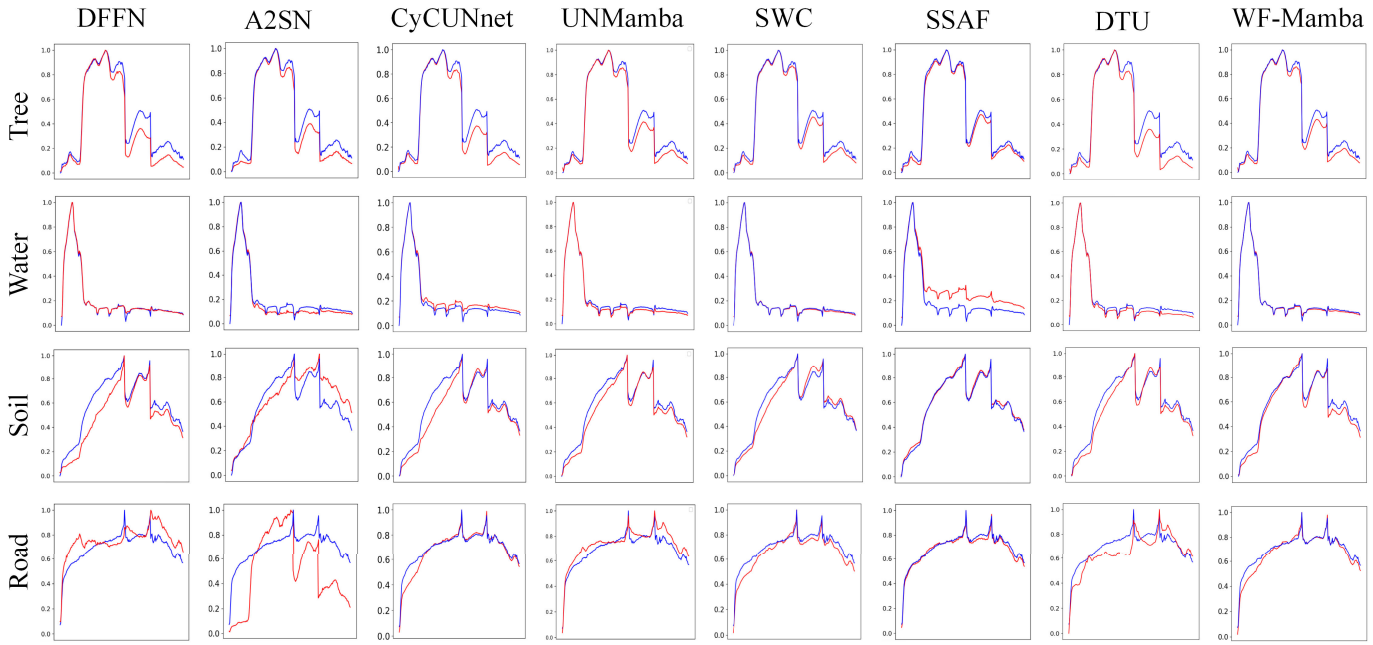


Fig. 11. Comparison of extracted endmember signatures on the Jasper Ridge dataset. The red lines represent the GT, while the blue lines denote the estimated signatures by different methods.

To comprehensively evaluate the robustness of the proposed WF-Mamba and validate our hyperparameter selection, we conducted a sensitivity analysis on the learning rate across three real-world datasets. The impact of varying learning rates on the SAD and RMSE metrics is illustrated in Fig. 12.

As shown in the parameter sensitivity curves, different spatial-spectral complexities necessitate adaptive learning rate configurations. For the Jasper Ridge dataset [Fig. 12(c)], the model exhibits a tradeoff between spatial reconstruction and spectral fidelity. While a lower learning rate yields marginally better SAD, the RMSE reaches a clear local minimum at 0.0009, indicating that this configuration achieves the optimal balance for accurately estimating fractional abundances in natural scenes.

For the highly complex Houston dataset [Fig. 12(a)], which is characterized by dense urban structures and severe shadow interference, the model's performance is highly sensitive to the learning rate. The metrics exhibit a sharp "V-shaped" trend, with both SAD and RMSE achieving optimal performance simultaneously at a learning rate of 0.005. Exceeding this value leads to a drastic degradation in RMSE, demonstrating that an excessive step size causes optimization oscillations in

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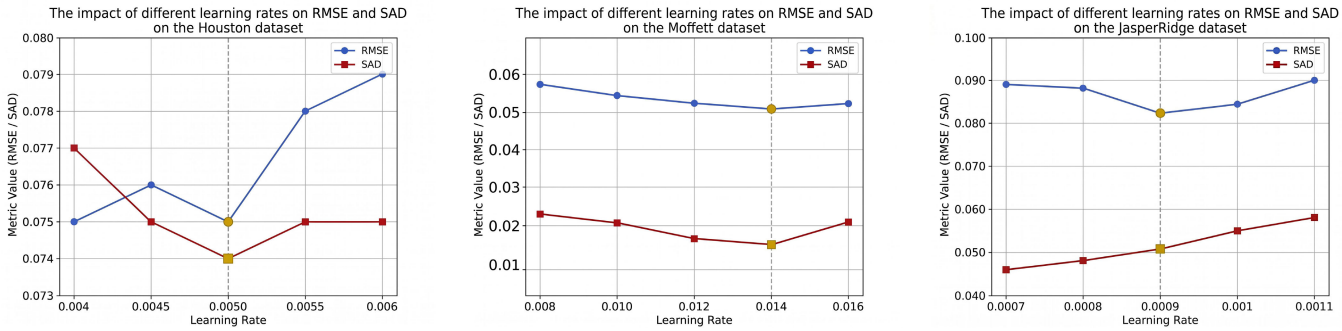


Fig. 12. Impact of different learning rates on the RMSE and SAD metrics across three real-world datasets. (a) Houston. (b) Moffett field. (c) Jasper Ridge. The optimal learning rate for each dataset is highlighted, demonstrating the model's convergence behavior under varying spatial–spectral complexities.

TABLE VII

ABLATION STUDY OF PROPOSED MODULES ON HOUSTON, MOFFETT FIELD, AND JASPER RIDGE DATASETS

SS	WFNO	ASF	Houston		Moffett		Jasper Ridge	
			RMSE (↓)	SAD (↓)	RMSE (↓)	SAD (↓)	RMSE (↓)	SAD (↓)
×	×	×	0.084	0.079	0.067	0.023	0.094	0.056
✓	×	×	0.081	0.079	0.056	0.021	0.090	0.056
×	✓	×	0.091	0.076	0.063	0.024	0.091	0.058
×	✓	✓	0.077	0.075	0.060	0.023	0.087	0.060
✓	✓	×	0.077	0.075	0.054	0.020	0.090	0.055
✓	✓	✓	**0.075**	**0.074**	**0.051**	**0.016**	**0.082**	**0.051**

the WFNO branch, preventing the model from capturing fine-grained urban details. Conversely, the Moffett field dataset [Fig. 12(b)], dominated by coastal zones and water surface reflections, requires a significantly higher learning rate to escape local optima. Both evaluation metrics show a smooth convex trajectory, converging perfectly at a learning rate of 0.014.

V. CONCLUSION

In this article, we proposed WF-Mamba, a novel dual-stream spatial-frequency SSM tailored for robust blind HU. By synergistically integrating the linear-complexity sequence modeling of Mamba with the global frequency-domain perception of the WFNO, this architecture effectively overcomes the spatial topology disruption and high-frequency noise susceptibility inherent in canonical 1-D SSMs. Extensive experiments on synthetic and complex real-world datasets demonstrate that WF-Mamba achieves highly competitive unmixing precision and robustness in both endmember extraction and abundance estimation while maintaining computational efficiency. Subsequent extensions of this framework will address complex nonlinear mixing effects caused by atmospheric scattering and explore lightweight module configurations to facilitate real-time, onboard unmixing for resource-constrained airborne platforms.

REFERENCES

[1] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[2] W.-K. Ma et al., "A signal processing perspective on hyperspectral unmixing: Insights from remote sensing," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 67–81, Jan. 2014.

[3] J. M. P. Nascimento and J. M. Bioucas-Dias, "Hyperspectral unmixing based on mixtures of Dirichlet components," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 863–878, Mar. 2012.

[4] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018.

[5] B. Koirala, B. Rasti, Z. Bnoulkacem, and P. Scheunders, "Nonlinear spectral unmixing using Bézier surfaces," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522416.

[6] K. E. Themelis, A. A. Rontogiannis, and K. D. Koutroumbas, "A novel hierarchical Bayesian approach for sparse semisupervised hyperspectral unmixing," *IEEE Trans. Signal Process.*, vol. 60, no. 2, pp. 585–599, Feb. 2012.

[7] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2014.

[8] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2013.

[9] S. Jia and Y. Qian, "Constrained nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 1, pp. 161–173, Jan. 2009.

[10] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.

[11] S. Ozkan, B. Kaya, and G. B. Akar, "EndNet: Sparse autoencoder network for endmember extraction and hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 482–496, Jan. 2019.

[12] J. Chen, P. Gamba, and J. Li, "MAHUM: A multitasks autoencoder hyperspectral unmixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519516.

[13] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

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[14] D. Hong et al., "Endmember-guided unmixing network (EGU-Net): A general deep learning framework for self-supervised hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6518–6531, Nov. 2022.
[15] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.
[16] T. Fang, F. Zhu, and J. Chen, "Hyperspectral unmixing based on multilinear mixing model using convolutional autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507316.
[17] M. Li, B. Yang, and B. Wang, "EMLM-Net: An extended multilinear mixing model-inspired dual-stream network for unsupervised nonlinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509116.
[18] Y. Su et al., "DAAN: A deep autoencoder-based augmented network for blind multilinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512715.
[19] X. Zhang, Y. Sun, J. Zhang, P. Wu, and L. Jiao, "Hyperspectral unmixing via deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1755–1759, Nov. 2018.
[20] M. Zhao, S. Shi, J. Chen, and N. Dobigeon, "A 3-D-CNN framework for hyperspectral unmixing with spectral variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521914.
[21] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral–spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2021.
[22] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.
[23] Z. Han, D. Hong, L. Gao, J. Yao, B. Zhang, and J. Chanussot, "Multimodal hyperspectral unmixing: Insights from attention networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524913.
[24] Z. Han, D. Hong, L. Gao, B. Zhang, M. Huang, and J. Chanussot, "AutoNAS: Automatic neural architecture search for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532214.
[25] Y. Su, L. Gao, A. Plaza, X. Sun, M. Jiang, and G. Yang, "SRViT: Self-supervised relation-aware vision transformer for hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 17585–17598, Oct. 2025.
[26] P. Ghosh, S. K. Roy, B. Koirala, B. Rasti, and P. Scheunders, "Hyperspectral unmixing using transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535116.
[27] Y. Duan, X. Xu, T. Li, B. Pan, and Z. Shi, "UnDAT: Double-aware transformer for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522012.
[28] C. Wang, J. Gao, F. Zhu, A. Halimi, and C. Richard, "DTU-Net: A multi-scale dilated transformer network for nonlinear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5502917.
[29] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.
[30] D. Chen, J. Zhang, and J. Li, "UNMamba: Cascaded spatial–spectral mamba for blind hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.
[31] P. Duan, S. Jin, X. Lu, L. Liang, X. Kang, and A. Plaza, "Domain-adaptive mamba for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 35, pp. 1206–1217, 2026.
[32] L. Liang et al., "DBMLLA: Double-branch mamba-like linear attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5524315.
[33] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.
[34] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.
[35] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," in *Proc. Int. Conf. Mach. Learn.*, 2024, pp. 1–14.
[36] M. Xu, J. Xu, S. Liu, H. Sheng, B. Shen, and K. Hou, "Stationary wavelet convolutional network with generative feature learning for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501613.
[37] D. Yin, R. G. Lopes, J. Shlens, E. D. Cubuk, and J. Gilmer, "A Fourier perspective on model robustness in computer vision," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–11.
[38] J. Guibas, M. Mardani, Z. Li, A. Tao, A. Anandkumar, and B. Catanzaro, "Adaptive Fourier neural operators: Efficient token mixers for transformers," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–15.
[39] P. Wang, Z. He, B. Huang, M. Dalla Mura, H. Leung, and J. Chanussot, "VOGTNet: Variational optimization-guided two-stage network for multispectral and panchromatic image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9268–9282, May 2025.
[40] Z. Li et al., "Fourier neural operator for parametric partial differential equations," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–16.
[41] P. Wang, Y. Tang, B. Qin, and D. Zhu, "A physics-guided reconstruction and injection framework with semantic consistency constraints for oriented SAR target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5104017.
[42] X. Liu and H. Tang, "DiffFNO: Diffusion Fourier neural operator," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 150–160.
[43] Y. Rao, W. Zhao, Z. Zhu, J. Lu, and J. Zhou, "Global filter networks for image classification," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 980–993.
[44] L. Liang et al., "S2PW-mamba: Pinwheel and wavelet-based spatial–spectral mamba for hyperspectral image classification," *Expert Syst. Appl.*, vol. 324, Aug. 2026, Art. no. 132497. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417426014107
[45] Y. Wang, H. Zhang, C. Lai, and X. Hu, "Transfer learning Fourier neural operator for solving parametric frequency-domain wave equations," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5923211.
[46] X. Tao, B. Koirala, A. Plaza, and P. Scheunders, "A new dual-feature fusion network for enhanced hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540113.
[47] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.
[48] W. Gao, J. Yang, Y. Zhang, Y. Akoudad, and J. Chen, "SSAF-Net: A spatial–spectral adaptive fusion network for hyperspectral unmixing with endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506015.

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